

Working the land

People have always been looking to come up with inventions that would make farming more efficient. If the greatest game-changer was the plow, then the second most significant innovation was the motor engine, which offered far more power than horses, and drove tractors, combine harvesters, and all manner of heavy farming machinery.

Threshing machine

- **What?** Steam-driven thresher
- **Who?** Andrew Meikle
- **Where and when?** UK, 1788

Farm laborers used to thresh wheat by hand, beating the harvested corn with sticks to separate the grain from the stalks and its outer covering, known as chaff. In 1788, Scottish millwright Andrew Meikle invented a machine that could do this quicker. It was powered by a separate steam engine.



1860 thresher

Steam-powered tractor

- **What?** Mobile steam engine
- **Who?** Charles Burrell
- **Where and when?** UK, 1856

In the 1790s, stationary steam engines were used on farms to power threshing machines. A self-moving steam engine was exhibited in 1842, but Englishman Charles Burrell built the first practical steam-powered tractor that could cope with rough farm terrain in 1856.

McCormick's reaper at work

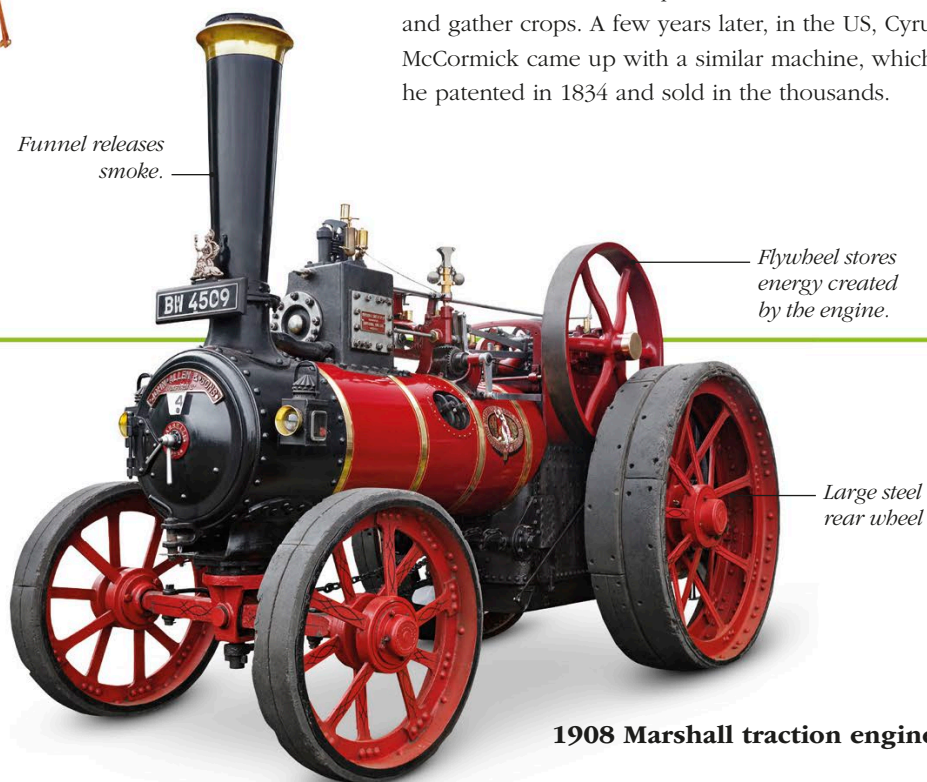


Mechanical reaping machine

- **What?** Horse-powered reaper
- **Who?** Patrick Bell
- **Where and when?** UK, 1827

Without mechanical help, harvesting requires many people. In 1826, Scottish farmer Patrick Bell invented a machine that could be pulled behind a horse to cut and gather crops. A few years later, in the US, Cyrus McCormick came up with a similar machine, which he patented in 1834 and sold in the thousands.

Funnel releases smoke.



Flywheel stores energy created by the engine.

Large steel rear wheel

1908 Marshall traction engine